

WASP-12b

R-1091

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_161218 · created 2026-07-30T17:34:33+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457740.67072 +/- 0.00034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.045
Transit depth (Rp/Rs)^2	2.09 +/- 1.29 [%]
Orbital Inclination (inc)	86.48 +/- 2.93
Airmass coefficient 1 (a1)	0.66 +/- 0.34
Airmass coefficient 2 (a2)	0.13 +/- 0.14
Scatter in the residuals of the lightcurve fit is	45.77 %
Best Comparison Star	#3 - [74, 155]
Optimal Aperture	2.17
Optimal Annulus	4.5
Transit Duration (day)	0.0894 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.144 ± 0.045 vs 0.1178 (deviation 0.58σ)
Duration fit vs published	0.0894 d vs 0.125 d (deviation 10.79σ)
Mid-time O–C	-0.6 min
Depth SNR	3.2
Residual scatter	45.77 %

Light curve

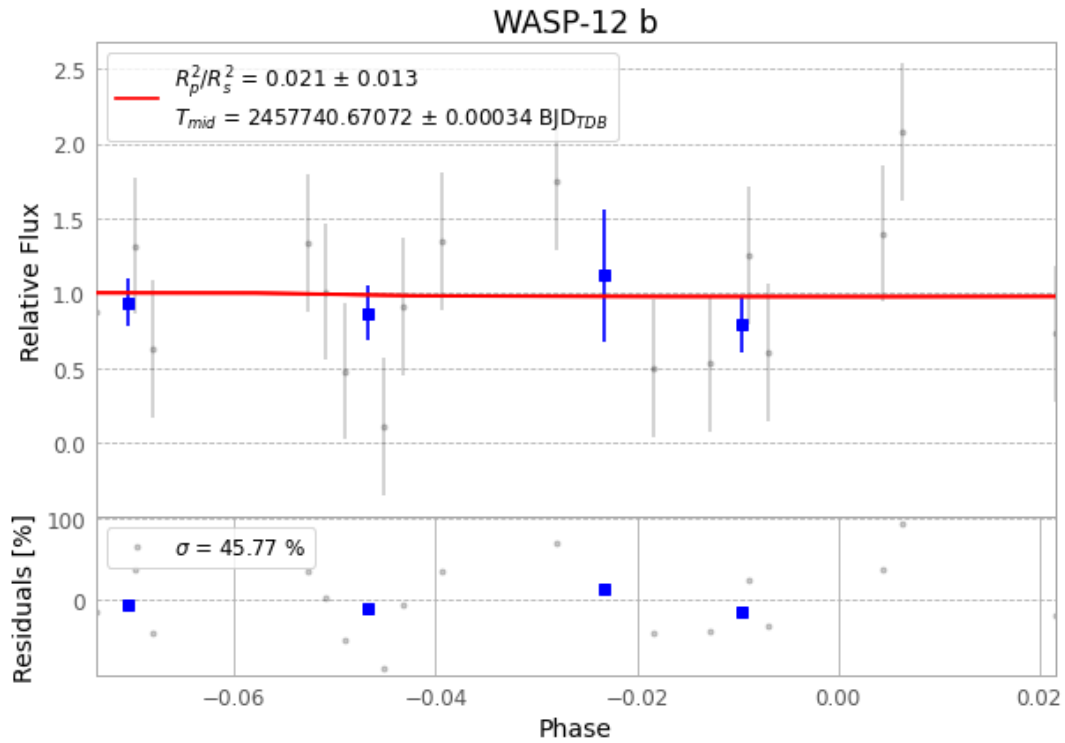


Figure 1 — FinalLightCurve_WASP-12 b_2016-12-17.png (SHA-256 631c82ea63a760ca...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [49, 401], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d9a2eaa46c5573ab...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

59 FITS frames (39 MB), WASP-12161218014512.FITS → WASP-12161218043912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-161218.log (e2d4f4626ec6...), FinalLightCurve_WASP-12 b_2016-12-17.png (631c82ea63a7...), FinalLightCurve_WASP-12 b_2016-12-17.csv (9f63465e56d8...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 17:34Z.